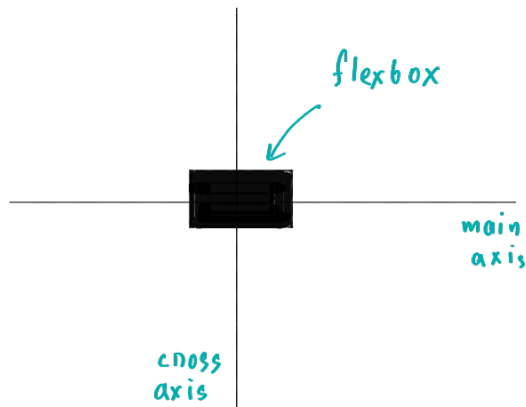


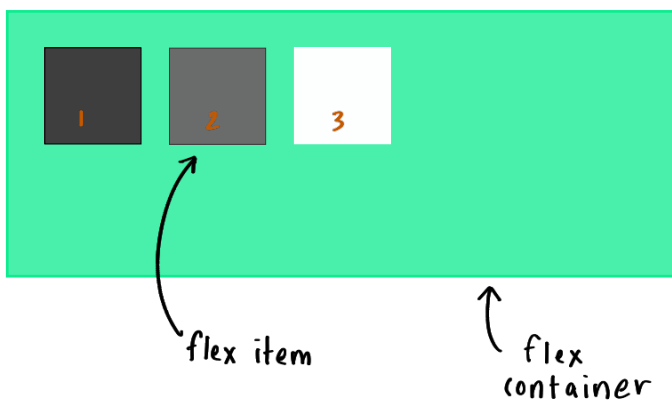
♥ FLEXBOX

- Flexible Box Layout
- 1D layout method for arranging items in rows or columns.
- Makes website **responsive**

The Flex Model



General Example



```
<div>  
  <div></div>  
  <div></div>  
  <div></div>  
</div>
```

```
display : flex ;
```

FLEXBOX DIRECTION

- Sets how flex items are placed in flex container, along which axis and direction

flexbox - direction : row ;

main axis, left to right

flexbox - direction : row - reverse ;

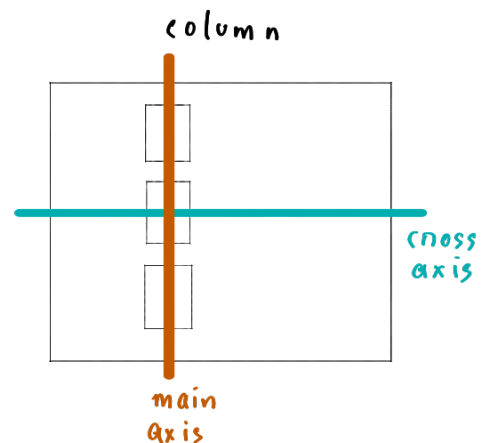
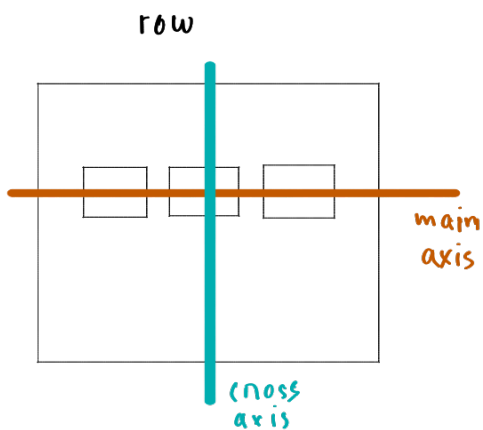
main axis, right to left

flexbox - direction : column ;

main axis, top to bottom

flexbox - direction : column - reverse ;

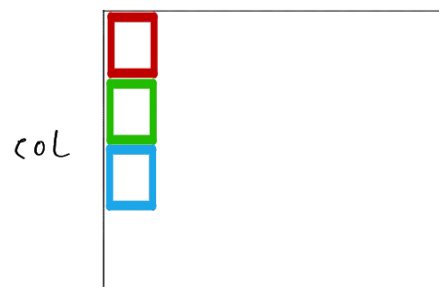
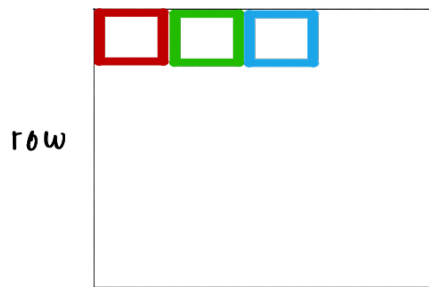
main axis, bottom to top



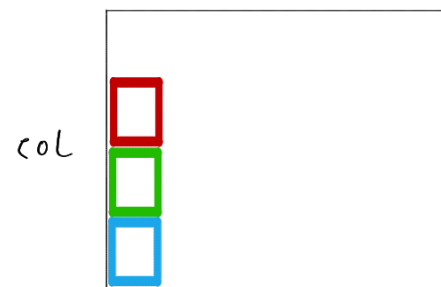
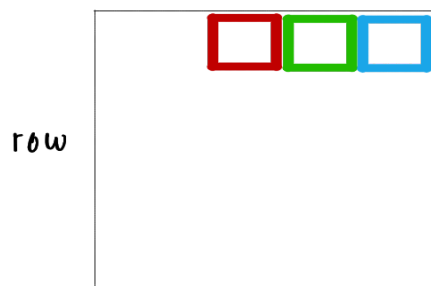
JUSTIFY CONTENT

– Tells how the browser distributes space between and around content items along the main axis

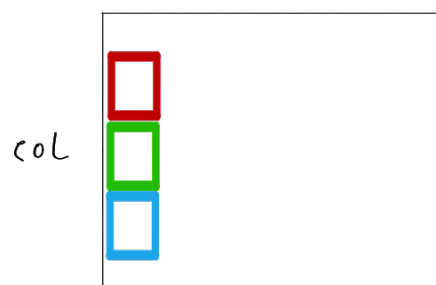
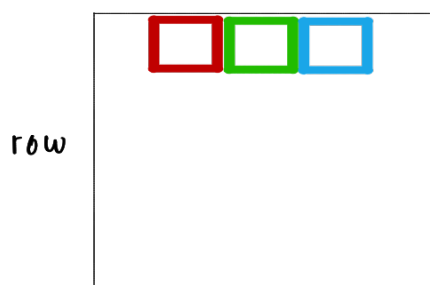
- `justify-content : flex-start ;`



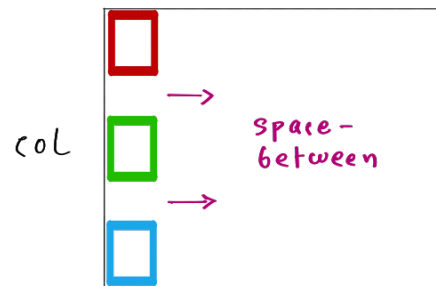
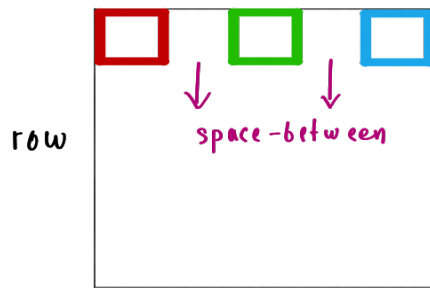
- `justify-content : flex-end ;`



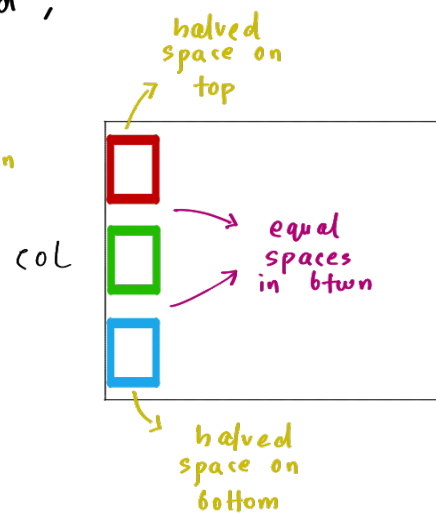
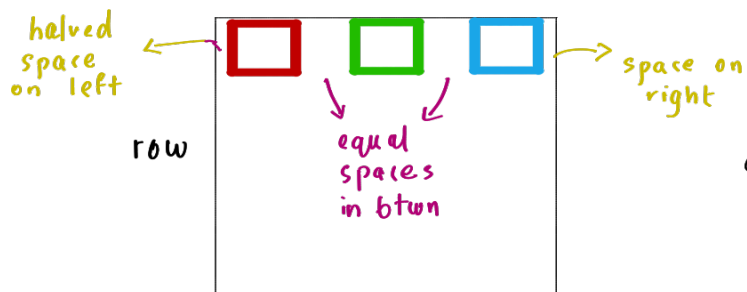
- `justify-content : center ;`



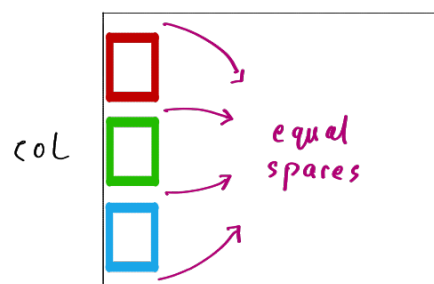
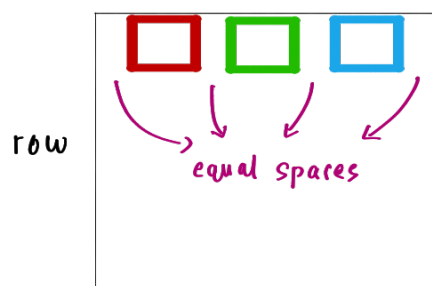
- justify - content : space - between ;



- justify - content : space - around ;



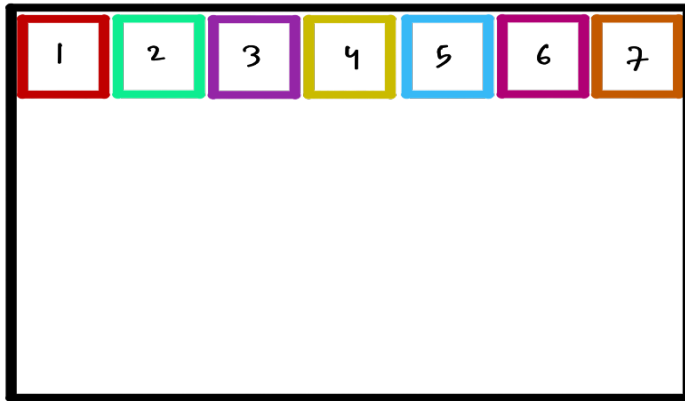
- justify - content : space - evenly ;



FLEX WRAP

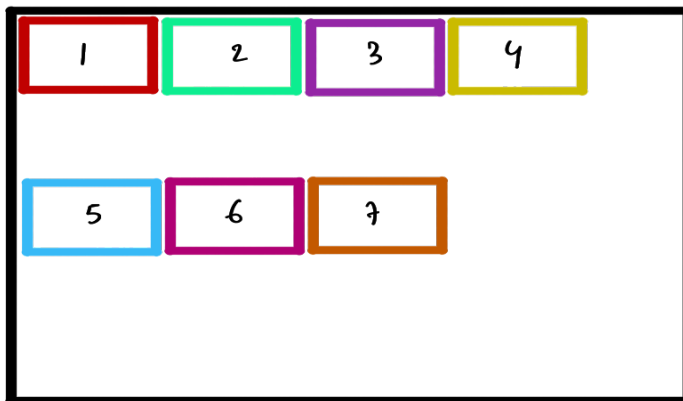
— See whether flex items are forced onto one line or can wrap onto multiple lines

- `flex-wrap : nowrap ;`



} — No wrapping
— elements dimension are manipulated to make them fit in one line.

- `flex-wrap : wrap ;`



} wrap

- `flex-wrap : wrap - reverse ;`



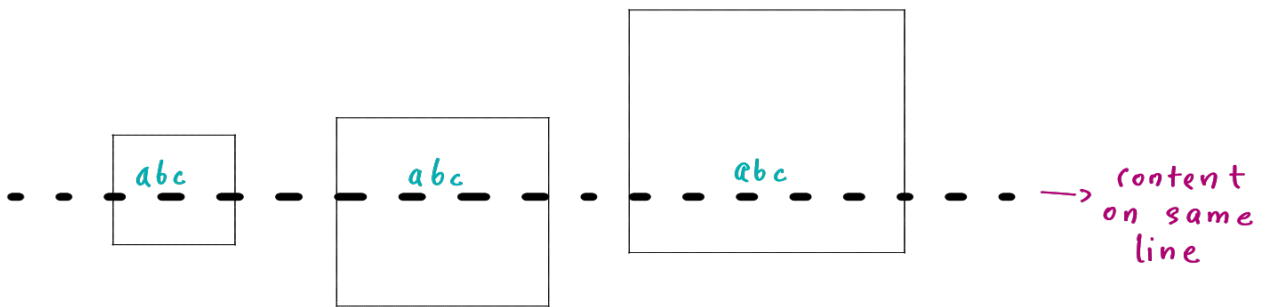
—> cross axis
reverse direction
wrap

ALIGN ITEMS

- Distributes our items along the cross axis

- `align-item : flex-start ;`
- `align-item : flex-end ;`
- `align-item : center ;`
- `align-item : baseline ;`

} same as justify content
but along cross axis



ALIGN CONTENT

→ like justify content

- Sets the distribution of space between and around content items along a flexbox's cross-axis

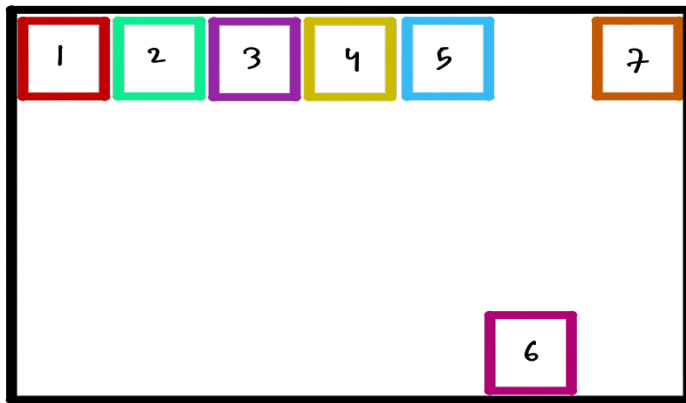
- `align-content: flex-start / flex-end / center ;`
- `align-content: space-between / space-around / space-evenly ;`
- `align-content: baseline ;`

ALIGN SELF

- Align an individual item along cross axis.

- `align-self: flex-start ;`
- `align-self: flex-end ;`
- `align-self: center ;`
- `align-self: baseline ;`

* Priority : $\begin{matrix} \text{align} \\ \text{self} \end{matrix} > \begin{matrix} \text{align} \\ \text{items} \end{matrix}$



```
# pink {  
  align-self: flex-end;  
}
```

FLEX SIZING

1 flex-basis

— It sets initial main size of a flex item

`flex-basis: 100px;` → only changes the size of individual item

2 flex-grow

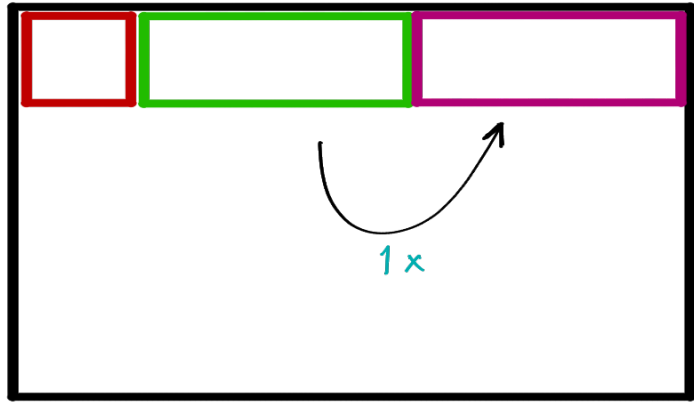
— It specifies how much of flex container's remaining space should be assigned to the flex item

`flex-grow: 1;` → default
↳ in proportion (1x, 2x, 3x, ...)
→ no unit

①

```
# green {  
  flex-grow : 1 ;  
}
```

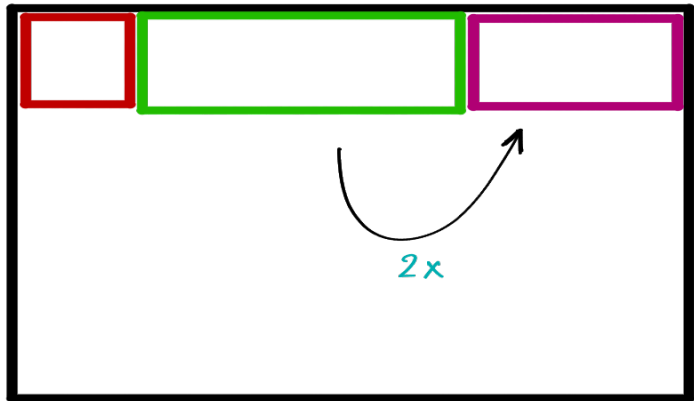
```
# pink {  
  flex-grow : 1 ;  
}
```



②

```
# green {  
  flex-grow : 2 ;  
}
```

```
# pink {  
  flex-grow : 1 ;  
}
```



③ flex-shrink (when item overflow outside container)

— It sets the flex shrink factor of a flex item.

```
flex-shrink : 1 ;
```

FLEX SHORTHAND

(1) flex — grow | shrink | basis

flex — 2 2 100px ;

(2) flex — grow | basis

flex — 2 100px ;

(3) flex — grow (unitless)

flex — 2 ;

(4) flex — basis

flex — 100px ;